

CoE 1199 Senior Project

ChessConnect

Final Testing and Analysis

Joseph Denk and Brendan Jugan

Department of Electrical and Computer Engineering



Testing Methodology

During the creation of ChessConnect, three phases of testing were used to ensure complete functionality: subsystem, integration, and final device. First, testing was conducted on each isolated subsystem to ensure that it operated as intended. The team then began testing subsystems that had been integrated together to perform complex functions used in the final device. Lastly, holistic testing on the completed device was used to confirm that full operation was achieved, refine the user experience, and detect any underlying bugs and issues.

Subsystem Testing

The core subsystems that makeup ChessConnect's architecture were constructed independently to speed up development time and act as building blocks when creating the higher functions of ChessConnect.

The internal gantry and control circuit were initially tested using a button setup for manual control at various speeds. Using four buttons for the cardinal directions and a fifth for activating the electromagnet, the project group was able to observe the maximum speeds and accuracy of movement as well as verify that electromagnet could maintain its control and strength throughout. This testing also revealed alignment issues and the fact that the rail system moved at half speed during diagonal movement which was later accounted for in software.

When creating the piece detection circuit, the group first needed to verify that the Reed switches could effectively be activated through the wooden material. This was accomplished via a basic circuit that used one of the magnetic switches to activate an LED. This gave the group vital information about the radius and consistency of activation and confirmed its viability as a solution. Once the Reed switches were mounted and wired into the four multiplexers and

microcontroller, test code was developed to scan all the switches to detect their open/closed state. The group confirmed individually that all 64 Reed switches could be triggered and detected in software using the designed system. This testing not only served as the basis for future move detection code but also ensured that all soldered connections were reliable and that pieces would not accidentally trigger neighboring sensors.

Lastly, API communication was tested by using small code segments as proof-of-concepts for the various communication requirements. The group first ensured that a basic TCP connection to Lichess.org could be established over WiFi. Throughout this process, the project group discovered early on that the original microcontroller and external WiFi module was incapable of connecting to the Lichess API and was able to quickly resolve this by choosing a different microcontroller. After a successful connection was established, the focus shifted towards obtaining and displaying a stream of an active game using API calls. This allowed the device to obtain and later interpret the live response of moves in an ongoing game.

Integration Testing

After subsystem testing was completed, integration testing was conducted to ensure that the subsystems interacted properly with each other to accomplish a larger task.

To test autonomous piece movement, the gantry was secured inside the enclosure and used to verify that moves from the online game stream were being correctly recognized and translated into physical movement. This test was also used to verify multiple aspects regarding the electromagnet and movement of pieces. During testing, the project group determined that the strength of the magnet was insufficient for moving pieces from within the chassis and purchased a new electromagnet with sufficient strength. It also revealed that diagonal movement was slightly inaccurate and could lead to a loss in calibration when repeatedly used. In response to

this, periodic recalibration was implemented based on the number of diagonal moves made during a game, resulting in better overall accuracy.

Move detection was tested by repeatedly making moves with the physical chess pieces and printing the source and destination squares to the LCD screen. This allowed the team to quickly verify both the functionality and consistency of the piece detection system. Additionally, the serial monitor was used to print the entire state of the board as pieces were moved to verify that the software scheme for internally storing the local game state of the device was functioning consistently. This revealed issues with the strength of the piece magnets in relation to Reed switches, oftentimes leading to very inconsistent detection. To remedy this issue, more tests were conducted on different magnet solutions to determine which provided the most consistent detection. This result was a design change from a single 8mm diameter magnet to two of these magnets side by side. This increased the radius of detection and provided a push and pull configuration which was much more consistent, however it also made movement very inaccurate because the point of magnetism on the pieces was no longer in the center. Ultimately, a design of three 8mm magnets stacked on top of each other offered the best detection while also allowing consistent piece movement.

This code was then combined with the API communication to test the software's ability to correctly format the various chess moves to be received by Lichess.org. This allowed the project team to learn how to send the various gameplay functions and inputs as well as handle edge case responses from the server.

Final Device Testing

Once the device construction and software were finalized, complete device testing was conducted to ensure that ChessConnect operated as designed.

Automatic piece movement was verified using active games on a personal Lichess.org account. This gave the project team manual control over what moves were received and made it easy to test special moves, edge cases, and movement time. Movement speeds were adjusted to optimize between fast operation and accurate piece handling and ensure that the automatic piece move time engineering specification was met.

Local gameplay on ChessConnect was used to confirm that all cases of legal and illegal moves could be detected and handled as well as measure the number of detection errors. Several complete chess games were played by third parties to accurately track instances of faulty detection due to inconsistent triggering. This also gave the team the opportunity to collect insights into potential improvements in usability. The measured number of piece detection errors was slightly higher than the threshold set during the design phase, an issue likely caused by the combination of board material and lower quality of the Reed switches. This combined with the user feedback led to the implementation of extra controls to correct and redo errors in move detection so that games can persist through these issues.

These tests were also used to monitor ChessConnect's internal record of the ongoing game over the microcontroller serial monitor and ensure that synchronization was being maintained with the online game even during special piece moves and edge cases. This allowed the team to fix several code bugs which would have otherwise gone unnoticed during the team's own preliminary testing.